

Program-6 : Deploy an Automated Build Pipeline using Docker Hub and GitHub Container Registry

Description :

This program demonstrates the creation of an automated container build and distribution pipeline by building a Docker image locally and publishing it to Docker Hub and GitHub Container Registry (GHCR).

Step-1 : Create Project Directory

```
cd ~  
mkdir program-6  
cd program-6
```

Step-2 : Create package.json

```
npm init -y
```

Step-3 : Create Dockerfile

```
nano Dockerfile
```

```
FROM node:18  
WORKDIR /app  
COPY package*.json ./  
RUN npm install  
COPY . .  
CMD ["node","index.js"]
```

Step-4 : Build Docker Image

```
sudo docker build -t program-6 .
```

Step-5 : login to Docker Hub

```
sudo docker login -u pavank1806  
Password : Docker Hub Personal Access Token
```

Step-6 : Tag Docker Image for Docker Hub

```
sudo docker tag program-6 pavank1806/program-6:latest
```

Step-7 : Push Image to Docker Hub

```
sudo docker push pavank1806/program-6:latest
```

```
pavan-k@pavank1RV24MC072:~/program-6$ sudo docker login -u pavank1806
```

 Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password:

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

```
pavan-k@pavank1RV24MC072:~/program-6$ sudo docker push pavank1806/program-6:latest
```

The push refers to repository [docker.io/pavank1806/program-6]

2daab2e5c515: Pushed

e23f099911d6: Pushed

885ee803cd87: Pushed

248497313b94: Pushed

79b2f47ad444: Pushed

45aa7001d0fc: Pushed

3e6b9d1a9511: Pushed

e21af5339ab2: Pushed

cda7f44f2bdd: Pushed

461077a72fb7: Pushed

c6b30c3f1696: Pushed

37927ed901b1: Pushed

3697be50c98b: Pushed

latest: digest: sha256:5cb333a3e9b088f4223c2e701383861ce4b1c323b9932decd01c187ac249d07e size: 856

Step-8 : Create GitHub Repository

Create a new repository on GitHub with name program-6

Step-9 : Initialize Git and Push Source Code

```
git init
```

```
git add .
```

```
git commit -m "Initial commit"
```

```
git branch -M main
```

```
git remote add origin https://github.com/Pavanbsc/program-6.git
```

```
git push -u origin main
```

```

pavan-k@pavank1RV24MC072:~/program-6$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/pavan-k/program-6/.git/
pavan-k@pavank1RV24MC072:~/program-6$ git branch -M main
pavan-k@pavank1RV24MC072:~/program-6$ git add .
git commit -m "Initial commit"
[main (root-commit) f3456f9] Initial commit
 2 files changed, 18 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 package.json
pavan-k@pavank1RV24MC072:~/program-6$ git config --global user.name "Pavanbsc"
pavan-k@pavank1RV24MC072:~/program-6$ git config --global user.email "pavankurubar9@gmail.com"
pavan-k@pavank1RV24MC072:~/program-6$ git remote add origin https://github.com/Pavanbsc/Program-6.git
pavan-k@pavank1RV24MC072:~/program-6$ git remote -v
origin  https://github.com/Pavanbsc/Program-6.git (fetch)
origin  https://github.com/Pavanbsc/Program-6.git (push)
pavan-k@pavank1RV24MC072:~/program-6$ git push -u origin main
Username for 'https://github.com': Pavanbsc
Password for 'https://Pavanbsc@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 510 bytes | 510.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Pavanbsc/Program-6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
pavan-k@pavank1RV24MC072:~/program-6$ sudo docker tag program-6 ghcr.io/pavanbsc/program-6:latest

```

Step-10 : Tag Image for GitHub Container Registry

`sudo docker tag program-6 ghcr.io/pavanbsc/program-6:latest`

Step-11 : Login to GitHub Container Registry

`echo GITHUB_PERSONAL_ACCESS_TOKEN | sudo docker login ghcr.io -u pavanbsc --password-stdin`

Step-12 : Push Image to GitHub Container Registry

`sudo docker push ghcr.io/pavanbsc/program-6:latest`

```

pavan-k@pavank1RV24MC072:~/program-6$ echo ghp_HZkXUd4VrWmcM8ar1GMpNpsUix035t4yI | sudo docker login ghcr.io -u pavanbsc --password-stdin

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
pavan-k@pavank1RV24MC072:~/program-6$ sudo docker push ghcr.io/pavanbsc/program-6:latest
The push refers to repository [ghcr.io/pavanbsc/program-6]
885ee803cd87: Pushed
3e6b9d1a9511: Pushed
461077a72fb7: Pushed
c6b30c3f1696: Pushed
3697be50c98b: Pushed
cda7f44f2bdd: Pushed
e23f099911d6: Pushed
e21af5339ab2: Pushed
248497313b94: Pushed
79b2f47ad444: Pushed
2daab2e5c515: Pushed
37927ed901b1: Pushed
45aa7001d0fc: Pushed
latest: digest: sha256:5cb333a3e9b088f4223c2e701383861ce4b1c323b9932dec01c187ac249d07e size: 856
pavan-k@pavank1RV24MC072:~/program-6$

```